Reminder

- Check each red section; these indicate the scores you can earn upon completing that part or deductions for specific reasons.
- Homework should be submitted before the announced deadline. Scores for **late** submissions will be **reduced by 20% per day.**
- If you encounter any problems or have questions, please post them on ee-class
- Plagiarism is strictly prohibited. If we find that your work is duplicated, you will receive **a score of 0.**
- We have tested the code on Gemini and other AI engines. Please do not provide us with **fully duplicated code and comments** from these sources. The score will be reduced by **50% per question** if you do so.

Question 1: Create your own FIR filters to filter audio signal (Total: 40%)

"HW2_Mix.wav" is a mix of 3 songs. Based on the ideal impulse responses given in slide **#x1** and the windowing functions in slide **#x2**, you need to design and apply different FIR filters in the time domain to separate the three audio signals from the given audio file. You can refer to the algorithm on slide **#x3**. Next, you are asked to reduce the sampling rates of signals and compare them. Finally, the output audio signals are too simple so you should apply one-fold echo and multiple-fold echo (slide **#x4**) to produce wonderful music.

Question 1-1 (5%):

Now, open an **editor** and write down the questions and your answers.

a. **What are the values of x1, x2, x3, and x4?** Alternatively, which additional resources or pages do you think could help you apply the concept to solve this problem?

   Hint: FIR Filter Design, Windowing Function, One-fold/Multiple-fold Echoes.

b. **Please explain the concepts on pages x1, x2, x3, and x4 or on the additional resources**. You need to highlight the important concepts in your paragraph. Use your own approach to show that you truly comprehend the concept, rather than simply copying and pasting from AI. The objective is to develop your ability to extract, distill, and clearly explain the knowledge.

   Hint: Highlight key information using structure, graphs, color, boldface, hand-drawn elements, or other methods.

Save the content to **Question1-1.pdf**. **(+5%)**

Question 1-2 (35% in total = 8% from the results + 27% from the code and comments).

Now, we are going to implement the code. Please open the **Homework_02_Question1_2_v2.ipynb**. The 8% score will be based on the results. In summary, each file accounts for 1%.

a. **Result**
   You need to store the audio results and include them in the submission: (total 8%)
   1. Store the three filtered audio files before reducing the sampling rates. **(+3%)**
   "Low_pass_[cutoff frequency].wav". "High_pass_[cutoff frequency].wav".
   "Band_pass_[cutoff frequency 1]_[cutoff frequency 2].wav".
   2. Store the three filtered audio files after reducing the sampling rates. **(+3%)**
   "Low_pass_[cutoff frequency]_2kHZ.wav". "High_pass_[cutoff frequency]_2kHZ.wav". "Band_pass_[cutoff frequency 1]_[cutoff frequency 2] _2kHZ.wav".
   3. Store two echo audio files and name "Echo_one.wav" and "Echo_multiple.wav". **(+2%)**

b. **Code (+27% in total, check each score listed in Homework_02_Question1_2_v2.ipynb)**
   The 27% is determined by how you implement the system. Please follow the instructions in the Jupyter Notebook and complete the parts marked as "**TODO**". Your code must display and output the results to **allow us to verify its correctness.**

   Add comments to everything you described in Question 1-1 to explain your code, ensuring the comments relate directly to your implementation, as TA all know AI can generate comments automatically.

   Key point: Comments are a good way to switch between projects and quickly recap ideas. Use your own words to highlight the code, don't be too long.

Reminder

- You **are not allowed to import any libraries** in your Jupyter Notebook file except those provided by us. If you do, you will receive a **penalty of 10%**, meaning the maximum score you can achieve is 17% (27% - 10%), even if your code functions correctly..
- Please compress your **Question1-1.pdf**, **Homework_02_Question1_2_v(your version).ipynb** and the **audio results** into a zip file named **HW2_Q1_(student ID)_(your name).zip**. Bonus: **+2%** for correct formatting. The total score remains the same if you have already achieved 100%. This encourages you to follow the rules of society.

Question 2: Motion estimation on a video sequence (Total 60%)

Write a program to perform motion estimation on a given video sequence. You are required to implement two search algorithms to find the motion vectors: the Full Search method and the 2D Logarithmic Search method. For both methods, estimate motion vectors for non-overlapping macroblocks within a search window of ±p pixels in both the horizontal and vertical directions. Use the Sum of Absolute Differences (SAD), as described in the class materials, as the metric for block matching.

Question 2-1 (5%):

Now, open an **editor** and explain the key concepts of the question.
Again, please do not **only** submit AI-generated, copy-and-paste text. You need to distill and understand the knowledge yourself. You can provide your **chain of thought (+3%)** along with the **final summary (+2%)**.

Useful question types include: What, Why, How, and Compare.

Question 2-2 (55% in total) :

Now, open **Homework_02_Question2-2_v1.ipynb**. Please follow the instructions in the Jupyter Notebook and complete the parts marked as "**TODO**". Your code must display and output the results to **allow us to verify its correctness.**

Try the two search ranges (p=8 and p=16) for two macroblock sizes (8x8 and 16x16) by using the full search and 2D logarithmic search methods. The reference image is 008.jpg, and the target image is 009.jpg.

a. Implement the full search algorithm **(code: +5%, describe the core logic: +3%)**
   i. Generate and show the predicted images by the full search algorithm for all the above combinations (2*2=4 images) **(+4%; each correct image adds 1%)**
   ii. Show the motion vectors images for all of the above combinations. The motion vector should be drawn on the target image. (2*2=4 images) **(+4%; each correct image adds 1%)**
   iii. Show the residual images (in RGB) for all the above combinations (2*2=4 images) **(+4%; each correct image adds 1%)**
b. Implement the 2D logarithmic algorithm **(code: +5%, describe the core logic: +3%)**
   i. Generate and show the predicted images by 2D logarithmic algorithm for all the above combinations (2*2=4 images) **(+4%; each correct image adds 1%)**
   ii. Show the motion vectors images for all of the above combinations. The motion vector should be drawn on the target image. (2*2=4 images) **(+4%; each correct image adds 1%))**
   iii. Show the residual images (in RGB) for all the above combinations (2*2=4 images) **(+4%; each correct image adds 1%)**

c. Implement PSNR **(+3%)**
d. Report
    i. Run the full search algorithm, compare and discuss the motion-based image prediction quality for all the settings **(+4%)**
    ii. Run the 2D logarithmic algorithm, compare and discuss the motion-based image prediction quality for all the settings **(+4%)**
    iii. Plot SAD and PSNR, Compare the SAD and PSNR of the full search method and the 2d logarithmic search **(+4%)**

Reminder

- Your code should work correctly and the generated results must be consistent to your results in report
- Please compress your code and result in a zip file named **HW2_Q2_(student id)_(your name).zip** and upload to ee-class.
- The zip contains **Homework_02_Question2-2_v(your version).ipynb** and all the **results** (predicted images, motion vector images, and residual images) mentioned in the problem. Bonus: **+2%** for correct formatting. The total score remains the same if you have already achieved 100%.